

Loan Approval Prediction

Data Science Internship – Alfido Tech

Submitted By:

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Task 2

Executive Summary

This project focuses on predicting loan approval status using machine learning techniques. The dataset contains applicant information such as income, education, credit history, and property area. Data preprocessing, visualization, feature encoding, and model building were performed. Logistic Regression and Decision Tree models were compared using multiple evaluation metrics. Logistic Regression achieved the best performance and is recommended for deployment.

Dataset Description

The dataset contains information about loan applicants and their loan approval status.

Features include:

- Gender
- Married
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term
- Credit History

- Property Area
- Loan Status

[3] ✓ 0s

df.head()

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

Next steps: [Generate code with df](#) [New interactive sheet](#)

[6] ✓ 0s

df.info()

```

... <class 'pandas.core.frame.DataFrame'>
Index: 614 entries, 0 to 613
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID               614 non-null   object
1   Gender                601 non-null   object
2   Married               611 non-null   object
3   Dependents            599 non-null   object
4   Education              614 non-null   object
5   Self_Employed         582 non-null   object
6   ApplicantIncome        614 non-null   int64
7   CoapplicantIncome      614 non-null   float64
8   LoanAmount             592 non-null   float64
9   Loan_Amount_Term       600 non-null   float64
10  Credit_History         564 non-null   float64
11  Property_Area          614 non-null   object
12  Loan_Status            614 non-null   object
dtypes: float64(4), int64(1), object(8)
memory usage: 67.2+ KB

```

[7]
✓ 0s

df.isnull().sum()



...

0

Loan_ID

0

Gender

13

Married

3

Dependents

15

Education

0

Self_Employed

32

ApplicantIncome

0

CoapplicantIncome

0

LoanAmount

22

Loan_Amount_Term

14

Credit_History

50

Property_Area

0

Loan_Status

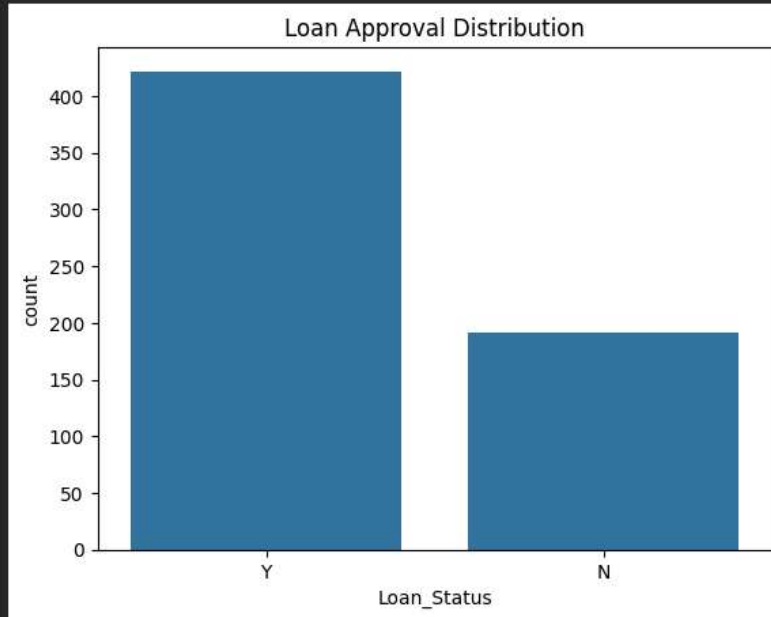
0

dtype: int64

[10]
✓ 1s

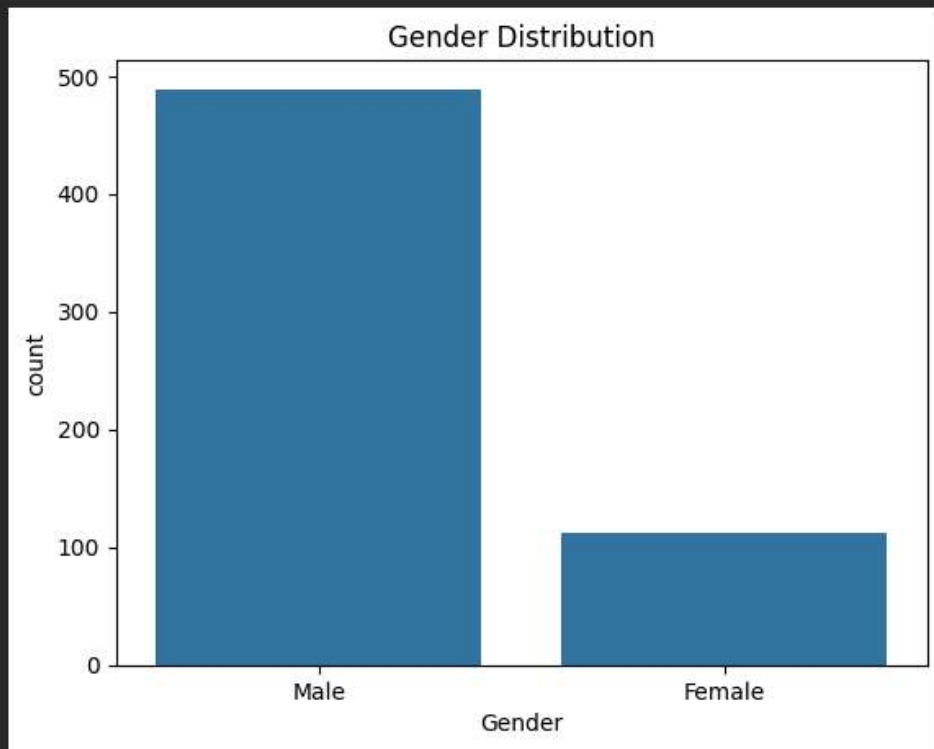
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Loan_Status', data=df)
plt.title('Loan Approval Distribution')
plt.show()
```



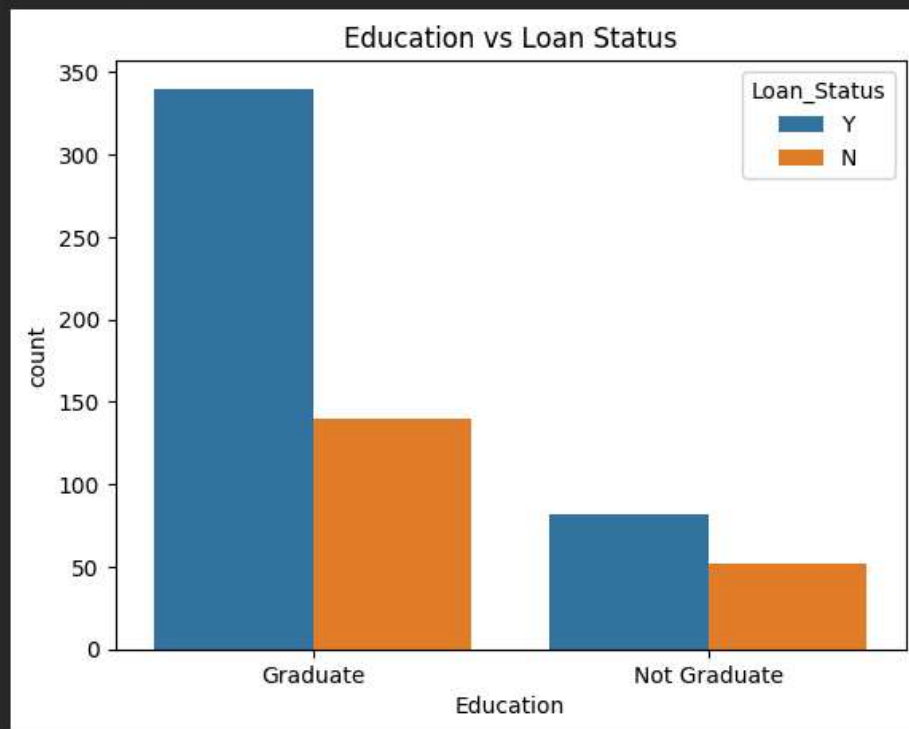
[11]
✓ 0s

```
sns.countplot(x='Gender', data=df)  
plt.title('Gender Distribution')  
plt.show()
```



[12]
✓ Os

```
sns.countplot(x='Education', hue='Loan_Status', data=df)  
plt.title('Education vs Loan Status')  
plt.show()
```





Logistic Regression Results

Accuracy: 78.86%

Precision: 75.96%

Recall: 98.75%

F1 Score: 85.87%

ROC-AUC Score: 73.87%


```
Accuracy: 0.7886178861788617  
Precision: 0.7596153846153846  
Recall: 0.9875  
F1 Score: 0.8586956521739131
```

```
from sklearn.metrics import roc_auc_score  
  
y_prob = model.predict_proba(X_test)[:,-1]  
  
print("ROC-AUC Score:", roc_auc_score(y_test, y_prob))  
  
ROC-AUC Score: 0.7386627906976744
```

Decision Tree Results

Accuracy: 69.11%

Precision: 75.61%

Recall: 77.50%

F1 Score: 76.54%

ROC-AUC Score: 65.49%

```
[34] 0s from sklearn.tree import DecisionTreeClassifier  
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score  
  
dt = DecisionTreeClassifier(random_state=42)  
  
dt.fit(X_train, y_train)  
  
dt_pred = dt.predict(X_test)  
  
print("Accuracy:", accuracy_score(y_test, dt_pred))  
print("Precision:", precision_score(y_test, dt_pred))  
print("Recall:", recall_score(y_test, dt_pred))  
print("F1 Score:", f1_score(y_test, dt_pred))  
  
dt_prob = dt.predict_proba(X_test)[:,-1]  
print("ROC-AUC:", roc_auc_score(y_test, dt_prob))  
  
... Accuracy: 0.6910569105691057  
Precision: 0.7560975609756098  
Recall: 0.775  
F1 Score: 0.7654320987654321  
ROC-AUC: 0.6549418604651164
```

Metric	Logistic Regression	Decision Tree
Accuracy	78.86%	69.11%
Precision	75.96%	75.61%
Recall	98.75%	77.50%
F1 Score	85.87%	76.54%
ROC-AUC	73.87%	65.49%

Business Interpretation

The Logistic Regression model outperformed the Decision Tree model across most evaluation metrics. The high recall score indicates that the model successfully identifies most eligible applicants for loan approval.

This can help financial institutions reduce the risk of rejecting deserving customers and improve operational efficiency by automating preliminary loan screening decisions.

Conclusion

A Loan Approval Prediction system was developed using machine learning techniques. After comparing Logistic Regression and Decision Tree models, Logistic Regression demonstrated superior performance with higher accuracy, recall, F1 Score, and ROC-AUC score.

Therefore, Logistic Regression is recommended as the final model for loan approval prediction and can assist financial institutions in making faster and more reliable lending decisions.